Lesson 6: Dendroecology — handout

Reading Environment and History through Tree Rings | 90 minutes | Draft v1 | 30 October 2026

Central question

Can a tree preserve evidence that insects ate its leaves? What survives in growth, anatomy, injuries and population ages? Explain dendroecological questions and distinguish observed changes from hypotheses about their causes.

Combine three types of evidence

1. Growth: timing, magnitude, duration and synchrony of suppression or release.

2. Injuries: crossdated scars, their likely causes and preservation.

3. Populations: establishment, regeneration and mortality, including missing individuals.

Interpreting causes

A canopy gap changes light and competition. But growth increases can also reflect climate, water availability, recovery from insect damage or management. A single wide ring or percentage change does not establish a cause. Responses may lag behind events.

Comparison stands and non-host species can help test hypotheses. Differences in species, site, size and management remain: comparison trees are not automatically experimental controls. Consider dead wood, stumps, regeneration, inventories, photographs and damage records as well.

Age and growth measures

A first preserved ring at sampling height in 1980 does not prove germination in 1980. Consider missing pith and the time needed to reach that height. Diameter alone is a poor age indicator, and surviving trees do not represent all past individuals.

For an ideal circular stem, basal area increment (BAI) = $\pi\,[(r + w)^2 - r^2]$, where r is previous radius and w is current ring width. With w = 1 mm, r = 10 mm adds about 66 mm², whereas r = 30 mm adds about 192 mm². Radius, stem form and sampling height introduce uncertainties. BAI alone does not determine biomass or carbon.

Exercise A: comparing growth changes

8 minutes: individual calculation 4; pair discussion 3; record conclusions 1.

All values are synthetic teaching data; ring widths are in mm. A, B and D represent a target stand; C a same-species comparison stand. An actual disturbance in 2000 is not supplied as a fact.

Year	A	B	C	D
1995	0.80	1.80	1.30	0.08
1996	0.90	1.90	1.40	0.09
1997	1.00	2.00	1.50	0.10
1998	1.10	2.10	1.60	0.11
1999	1.20	2.20	1.70	0.12
2000	1.80	2.70	1.20	0.18
2001	1.90	2.90	1.40	0.19
2002	2.00	3.00	1.50	0.20
2003	2.10	3.10	1.60	0.21
2004	2.20	3.30	1.80	0.22

Calculations and questions

1. Calculate M1 for 1995–1999 and M2 for 2000–2004. The supplied CSV can be used.

2. Calculate percent change G = (M2 − M1) / M1 × 100%, and absolute change M2 − M1 in mm.

3. Compare A and D using both measures. Do these establish a canopy gap?

4. Propose one additional observation to compare causal hypotheses.

The slide also supplies the period means. Calculate them first if useful. M1 must be positive; missing values are not zeros. These fixed five-year windows are a teaching choice, not universal release criteria. This exercise does not reproduce the ten-year averaging method of Nowacki and Abrams (1997).

Exercise B: scar counts and denominators

8 minutes: individual 4; pair discussion 3; record 1. All examples are fictional.

1 = fire scar recorded; 0 = recording but no scar; — = not recording. Recording eligibility is supplied as a teaching condition. In real studies, being alive alone does not establish that a tree could record a detectable scar.

Tree	1900	1920	1940	1960
T1	1	0	1	1
T2	—	1	1	0
T3	—	—	1	1
T4	—	—	—	1

Calculations and questions

1. Count recording trees and scarred trees for each year.

2. Calculate percentage = scarred trees / recording trees × 100%.

3. Compare 1940 and 1960. Does an equal scar count imply equal percentages?

4. Can this table establish fire size, severity or periodicity?

Limits of interpretation

Do not substitute zero for not recording. Even an eligible tree need not be scarred by a fire. The table contains selected fictional dates: intervals between them do not necessarily represent all fires. Scarred-tree percentages in this sample are not burned-area percentages or fire severity.

Explore actual records

NOAA's International Multiproxy Paleofire Database (IMPD; R37) includes different fire-history evidence, including scars and charcoal. Check recording-period and code definitions before analysis. R38 provides giant-sequoia examples; R40 combines different evidence for postfire forest development.

Terms and brief explanations

Term	Explanation
dendroecology	Studying ecological processes and forest history with dated rings.
stand dynamics	Changes in growth, recruitment and mortality within a stand.
canopy gap	An opening in the forest canopy.
disturbance	An event that changes ecosystem structure or resource conditions.
growth suppression	A period of relatively slow growth.
growth release	A sustained growth increase; operational criteria must be stated.
basal area increment (BAI)	Annual increase in stem cross-sectional area, not biomass itself.
regeneration	Forest renewal involving the appearance and growth of young trees.
establishment	A tree becoming established; distinct from the first ring at sampling height.
survivorship bias	Bias caused by considering survivors alone.
fire scar	Evidence left by fire damage to cambium and subsequent growth.
recording tree	A tree considered capable of recording and preserving a scar at that time.
defoliation	Loss of foliage, for example through insect feeding.
dendroentomology	Using tree rings to investigate insect histories.
intra-annual density fluctuations (IADFs)	Density or anatomical variations within a ring; not uniquely diagnostic of insects.

A scar-recording period, the measured ring interval and the tree's lifetime are not necessarily identical. Separate what can be observed from what actually occurred.

Research examples, links and assignment

Read insect evidence in widths and anatomy

Khasanov and Sandlersky (2018; R42) compared 49 oaks with 21 non-host ashes, examining intra-annual density fluctuations (IADFs) and abrupt growth changes. Ask what anatomy and the non-host comparison add beyond narrow rings alone. IADFs are not uniquely diagnostic of insect damage. The lesson uses a study-design summary; anatomical images will be selected later.

Other research examples

Cockchafer-related periodic narrow rings can complicate crossdating (R41). Check anatomy, damage records and independent dating evidence instead of relying on periodic resemblance. The slide's index series is synthetic.

Hirano and Morimoto (1999; R39) compared polluted and unpolluted Japanese-black-pine stands. Read how a climate model and SO_2 records support their interpretation.

Further study

- R35 Radial-growth averaging and disturbance history — Nowacki & Abrams (1997)
- R36 Release criteria and prior growth — Black & Abrams (2003)
- R37 Fire-history data / IMPD — NOAA NCEI
- R38 Giant sequoias and fire — US National Park Service
- R39 Japanese black pine growth and air pollution — Hirano & Morimoto (1999)
- R40 Fire history and postfire forest development — Fastie et al. (2002)
- R41 Cockchafer outbreaks and periodic narrow oak rings — Dendrochronologia (2013)
- R42 Does insect induced defoliation affect anatomical structure of oak wood? — Khasanov & Sandlersky (2018)

Link to the final assignment

State the question, predicted evidence, comparisons, sampling, crossdating and analysis, alternatives and additional evidence. Consult the syllabus requirement of 2–4 A4 pages and at least three references. Review a conclusion and its limits in approximately 200 Japanese characters or a comparable short English summary, add one reference and update the research outline. Next: dendroclimatology.

Further cases: survivors, eruptions and earthquakes

Several events in one pine

The USGS specimen account (R43) combines scars, staining and death in Pinus albicaulis. Consider drought, bark beetles and fungi together; the ultimate single cause of death remains unresolved. Follow the original image link; a teaching-image replacement will be considered later.

Hemlocks that survived infestation

Walker et al. (2014; R45) studied surviving trees at six sites: three declined, while the others showed no change or increased growth. The Virginia Tech story (R44) leads to the primary study. Discuss survivorship bias and alternatives such as changing competition.

Mount St. Helens

Use the independently dated 18 May 1980 eruption to examine spatial damage and survival, then suppression, scars, mortality and establishment. Dendrovolcanology combines tree evidence with deposits and topography. Distinguish this known event from reconstruction of earlier eruption dates.

The San Andreas fault

At Wrightwood, suppression from 1813 was linked with the late-1812 earthquake (R47–R49). Dendroseismology compares off-fault trees, sediments and historical records. Separate first growth response from event timing. This connects the lesson to dendrogeomorphology.

Further reading

- R43 A whitebark pine cross section: fire, insects and death — USGS (2023)
- R44 Eastern hemlocks surviving insect infestation — Virginia Tech (2014)
- R45 Radial growth changes following hemlock woolly adelgid infestation — Walker et al. (2014)
- R46 Mount St. Helens: a living laboratory — US Forest Service
- R47 Irregular recurrence of large earthquakes: evidence from trees — Jacoby et al. (1988)
- R48 The 1812 Wrightwood earthquake: tree rings and other evidence — Caltech SCEDC
- R49 Wrightwood tree-ring evidence and comparison trees — USGS Open-File Report 87-673
- R50 Reading the Tree Rings: a photo essay — The Nature Conservancy (2026)

Photo-reading task: choose one image in R50 and distinguish visible features, explanatory hypotheses and primary evidence you would check. Photographs are linked, not reproduced.